Compositional Taylor expansion in additive and non-additive models of Linear Logic

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This thesis is about

denotational semantics

```
def pow_two(n : int) -> int :
    res = 1
    for i in range(n)
        res = 2 * res
    return res
```

Reading code is tiring...

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def pow_two(n : int) -> int :
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Specification is better!

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Programming is **modular**

From specification to semantics

Specification is based on **words**. Not always enough.

- Critical software: Plane pilot
- Design of programming languages: Ocaml, Haskel, Rust
- Mix programming languages with maths: automated differentiation, probabilistic programming.

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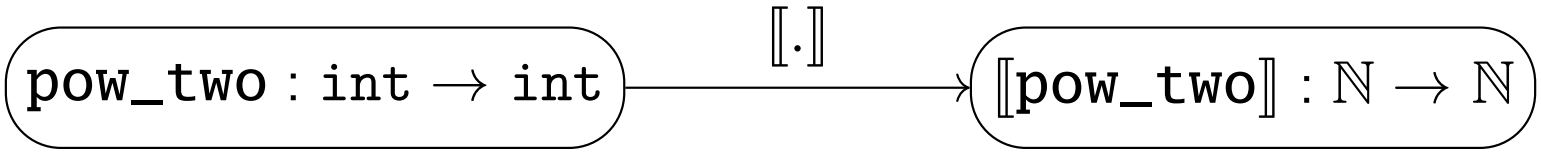
Semantics: specification based on **maths**

Denotational semantics

Denotational semantics: interpret programs as functions.



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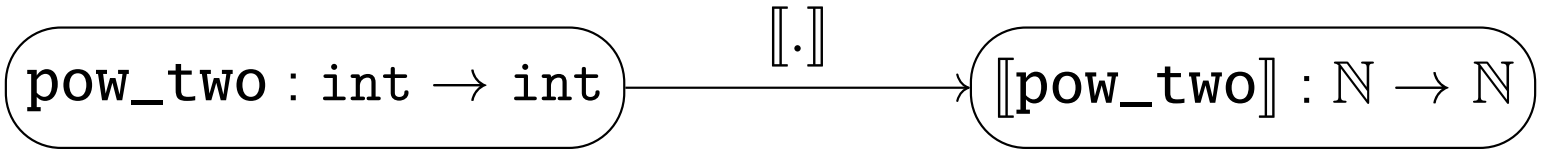

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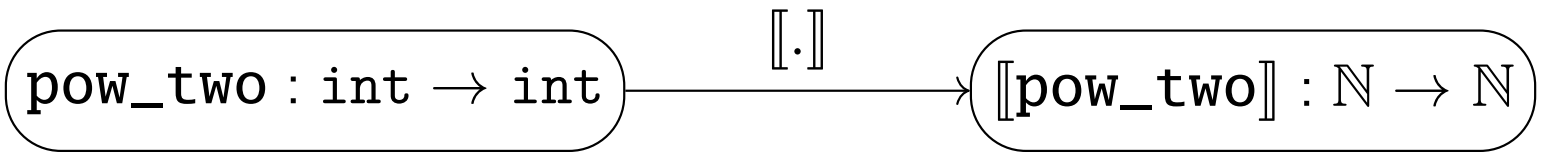

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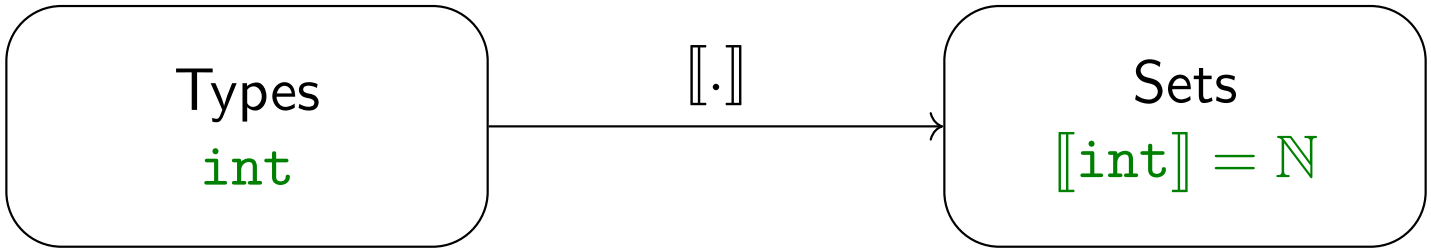

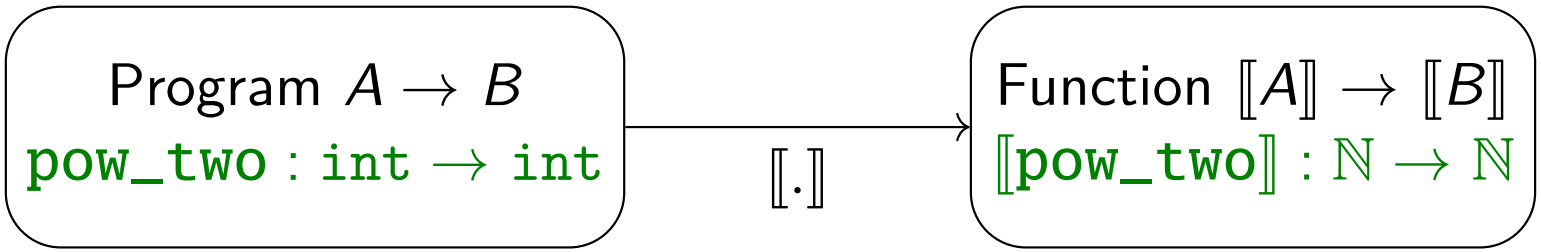

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Compositionality

$$[\![\texttt{pow\_four}]\!] = f \circ [\![\texttt{pow\_two}]\!]$$

with $f(m) = m \times m$

We need a mathematical theory

of composition

Category theory!!!!!

A category is

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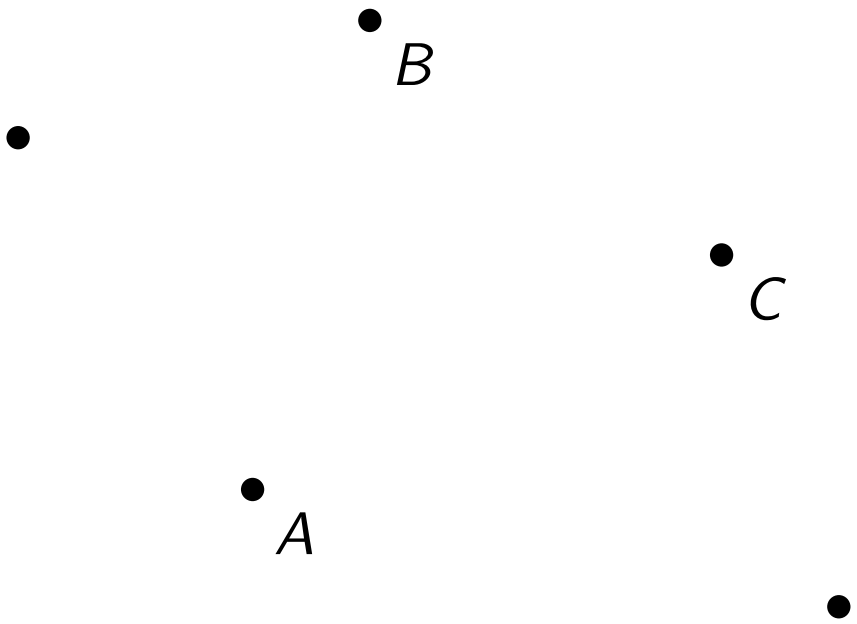
- Objects $A, B, C \ldots$



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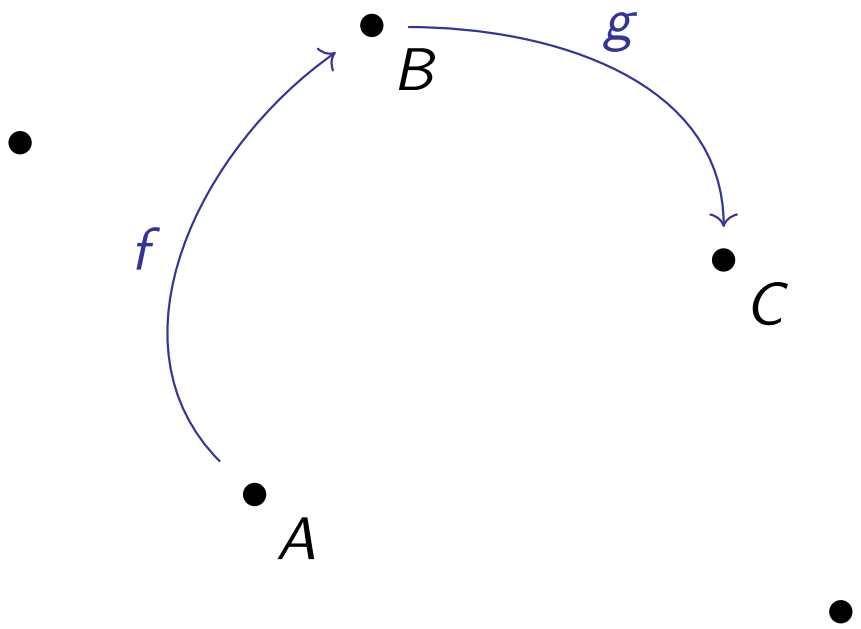

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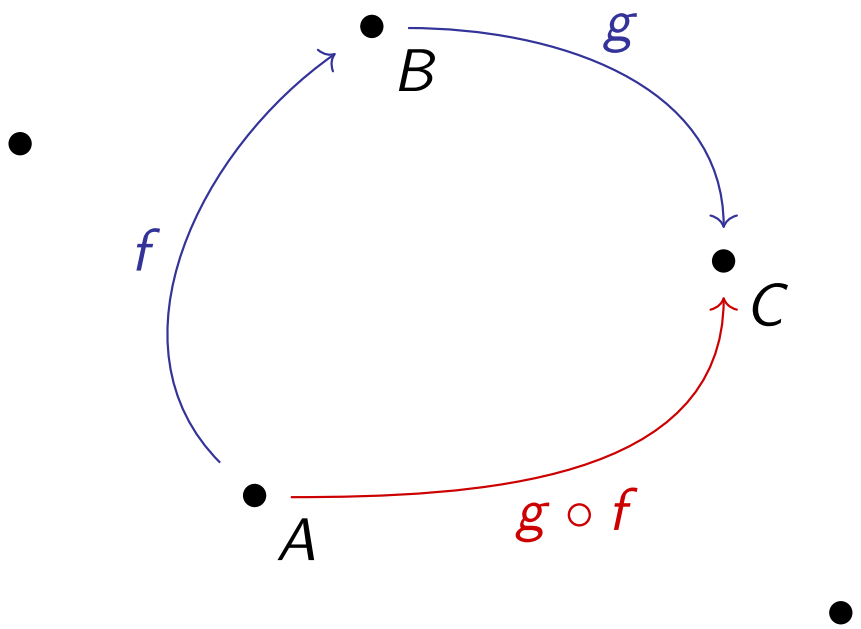

Category theory!!!!!

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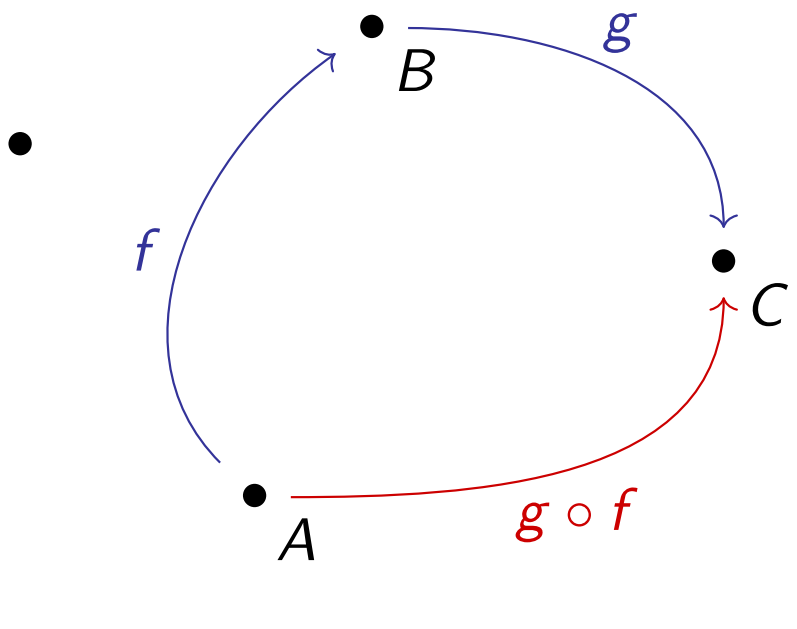
- Objects $A, B, C \ldots$ Sets
- Morphisms $f, g, h \ldots$ Functions
- Composition $\_ \circ \_$ Function composition



Syntax	Semantics in category of sets and functions	Generalization to other categories
Types	Sets	???
`int`	$\mathbb{N}$	
Program $A \to B$	Function $A \to B$	
`pow_two`	$[\![\texttt{pow\_two}]\!]$	???
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Product type	Cartesian product	???
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Product type `int` $\times$ `int`	Cartesian product $\mathbb{N} \times \mathbb{N}$	**Categorical product** N $\times$ N
Function type `int` $\to$ `int`	Set of functions $\mathbb{N} \to \mathbb{N}$	???

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Cartesian closed category

Category + Product + Internal Hom

Product type	Cartesian product	**Categorical product**
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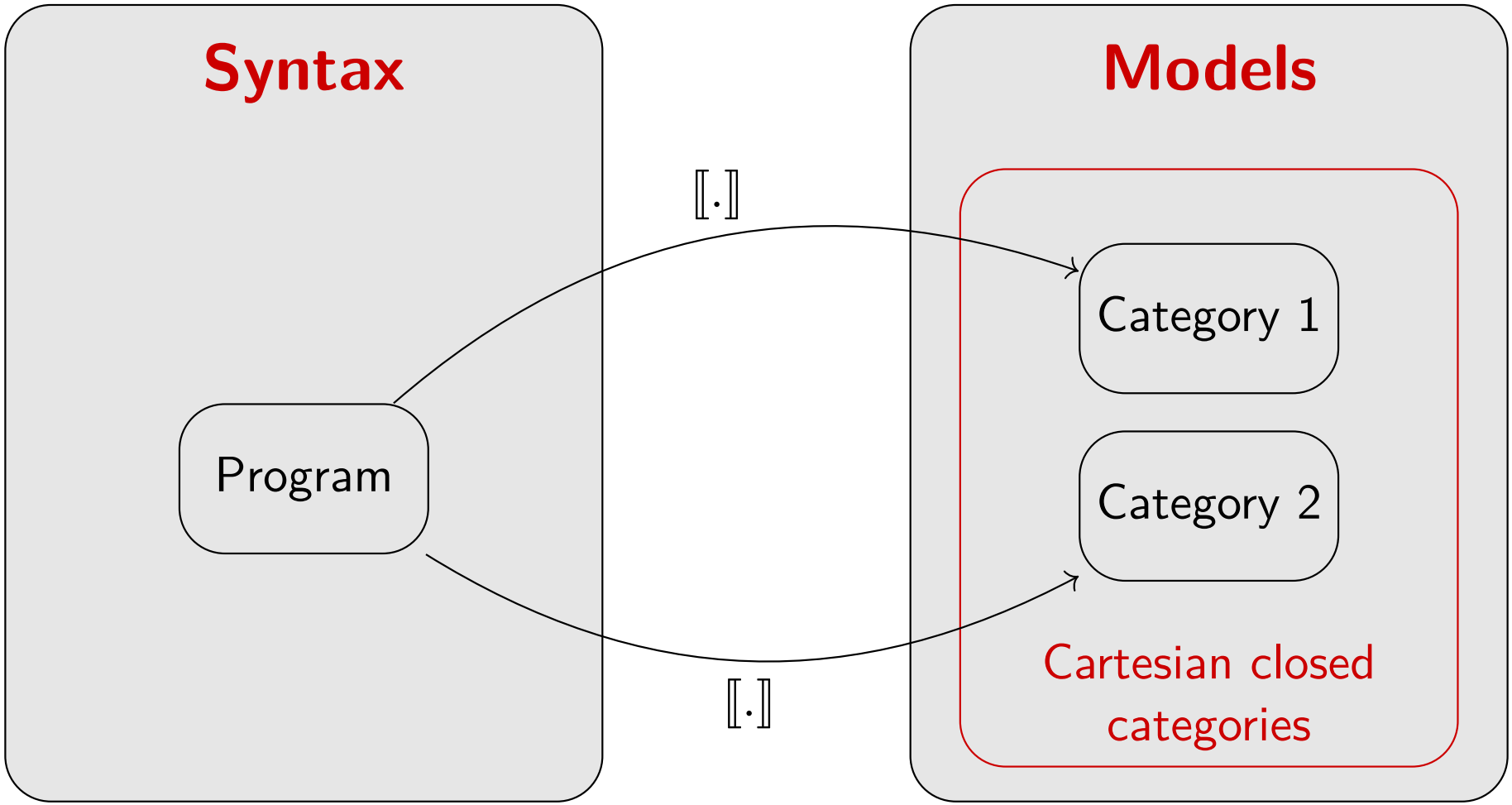
A recipe for models



Compositional Taylor expansion in additive and non-additive **models of Linear Logic**

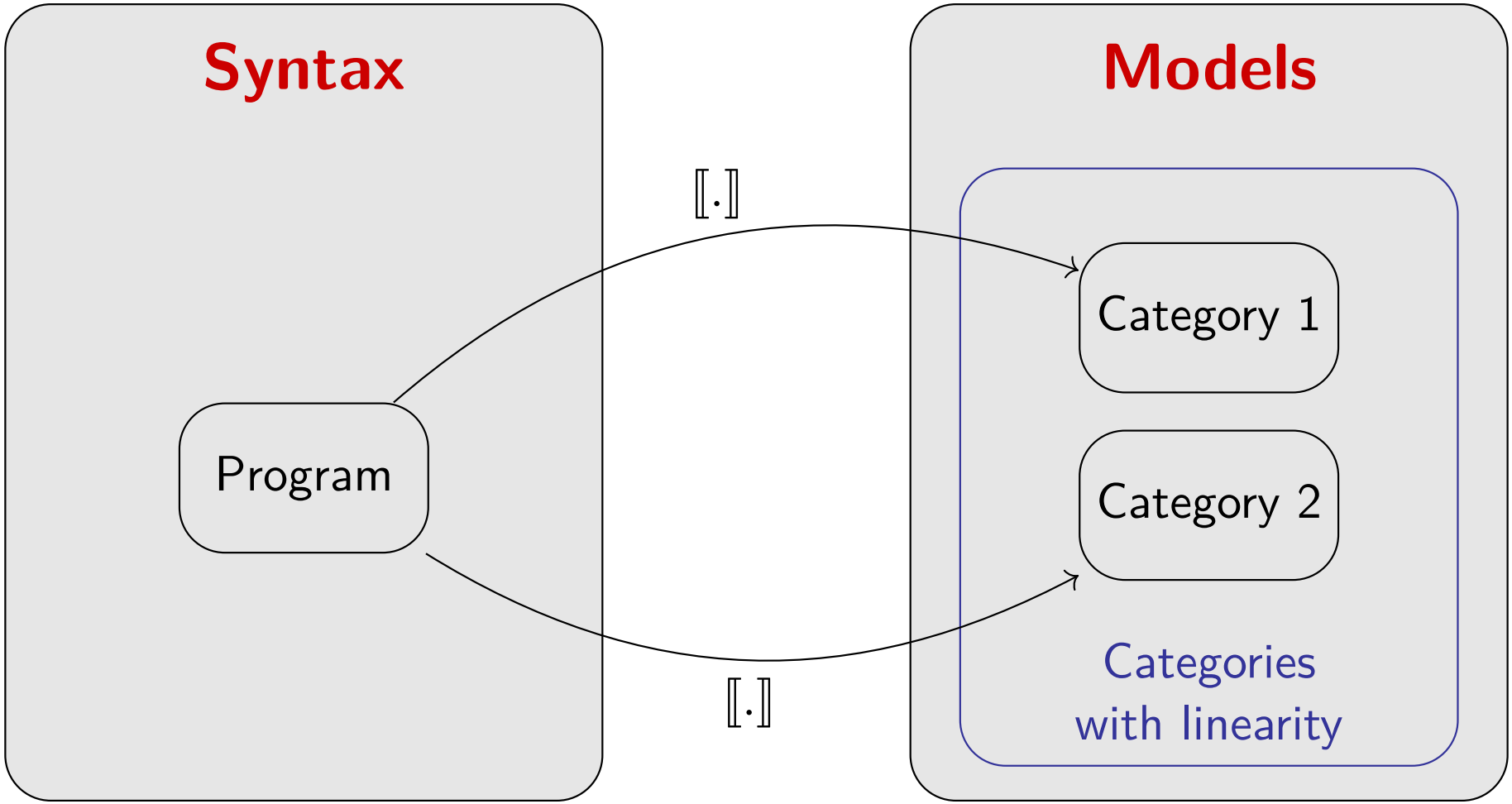
Linear Logic

Girard 1987 observed in some cartesian closed categories a **decomposition**

$$A \rightarrowtail B \quad = \quad !A \multimap B$$

- $\_ \multimap \_$: object that describes **linear** morphisms;
- $!\_$ is a repetition operator.

Reverse denotational semantics



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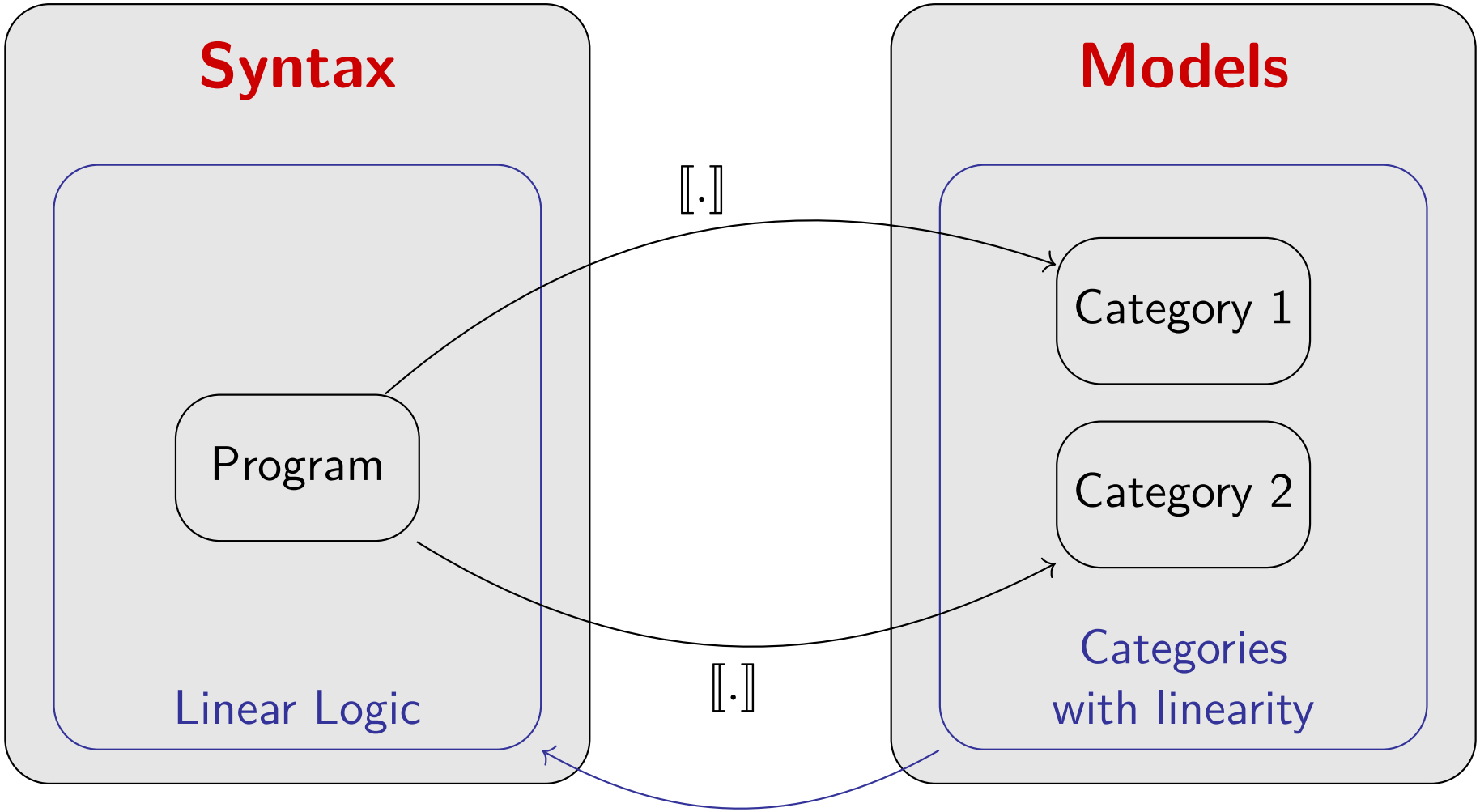

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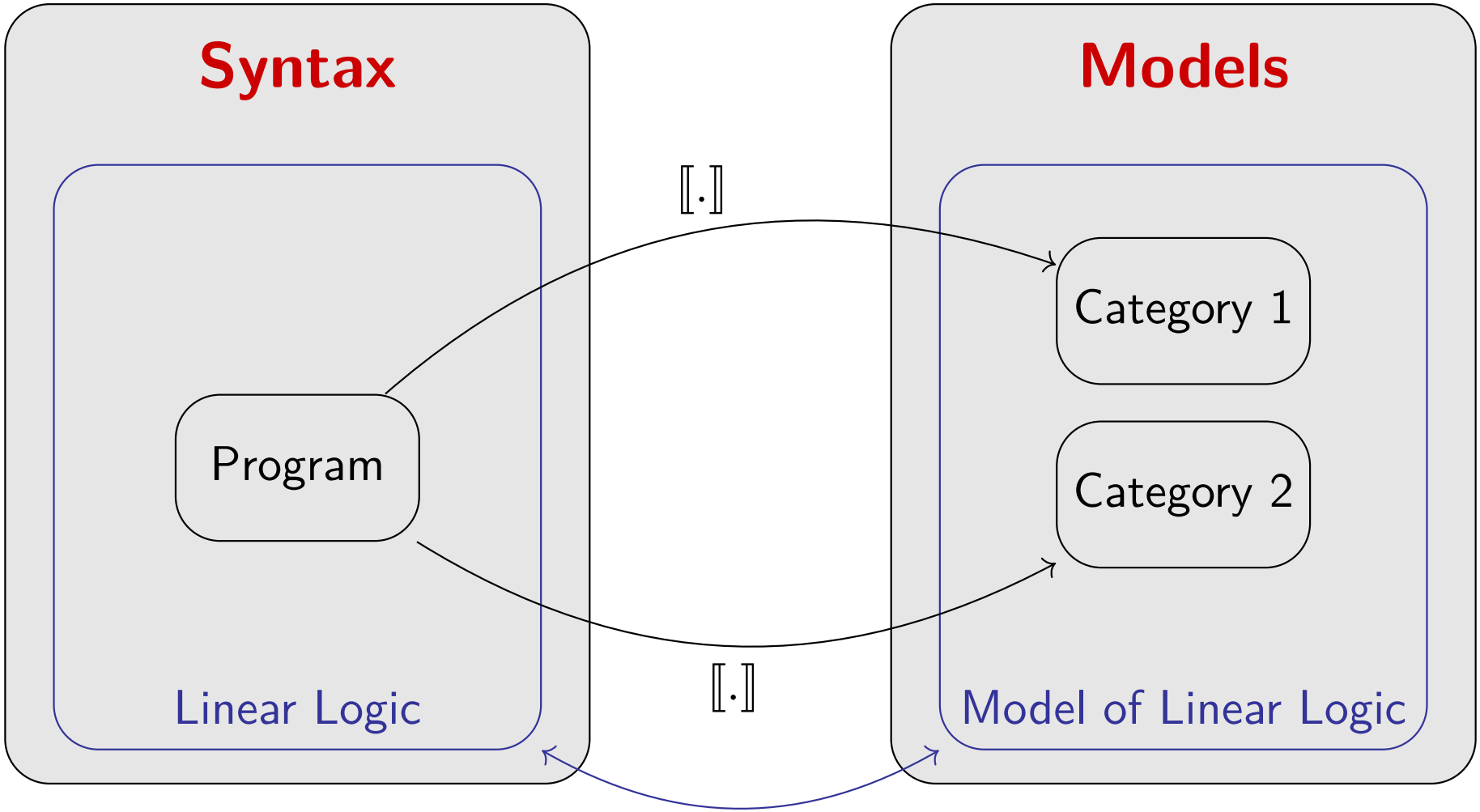

Linear Logic: between syntax and semantics

Syntax

- Linear programs $A \multimap B$
  Consumes exactly one A during computation
- Exponential type: $!A$
  Makes elements of A duplicable and erasable

- General programs:
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- Morphism in $\mathcal{L}(A, B)$
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- Comonad $!$ on $\mathcal{L}$
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- Kleisli category $\mathcal{L}_!$
  $\mathcal{L}_!(A, B) = \mathcal{L}(!A, B)$

Linear Logic: between syntax and semantics

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▶ Linear programs $A \multimap B$ Consumes exactly one A during computation	▶ Morphism in $\mathcal{L}(A, B)$ $\mathcal{L}$ is a linear category
▶ Exponential type: $!A$ Makes elements of A duplicable and erasable	▶ Comonad ! on $\mathcal{L}$! is a resource comonad
▶ General programs: Type $!A \multimap B$	▶ Kleisli category $\mathcal{L}_!$ $\mathcal{L}_!(A, B) = \mathcal{L}(!A, B)$

The Kleisli category $\mathcal{L}_!$ of the resource comonad $!\_$ is a cartesian closed category.

In many models of LL: finiteness spaces, Köthe spaces

- morphisms in $\mathcal{L}(X, Y)$: linear maps between vector spaces
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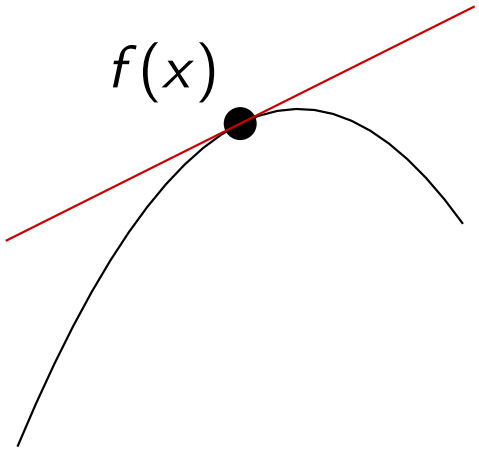
Differential λ-calculus and categorical differentiation

Differential calculus

A function $f : \mathbb{R} \to \mathbb{R}$ is derivable at $x \in \mathbb{R}$ if

$$f(x+u) = f(x) + f'(x) \cdot u + O(u^2)$$



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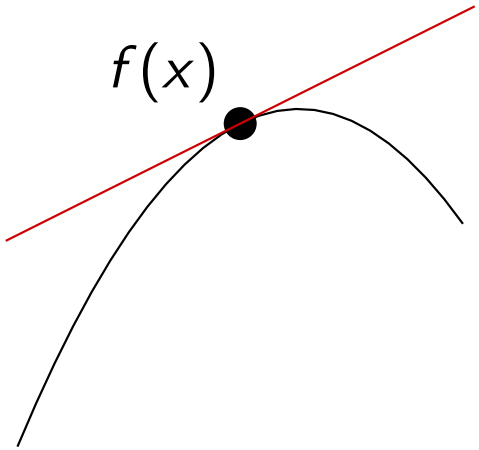

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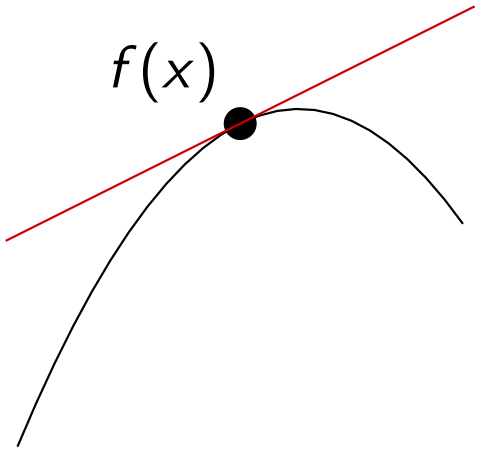

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Degree n Taylor expansion.

$$\boxed{f(x+u) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x) \cdot u^k + O(u^{n+1})}$$

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A function $f : E \to F$ is derivable at $x \in E$ if

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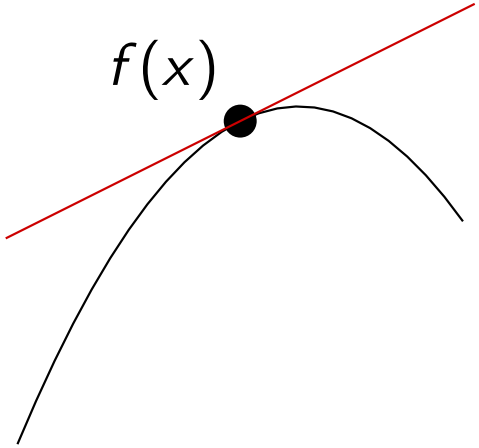

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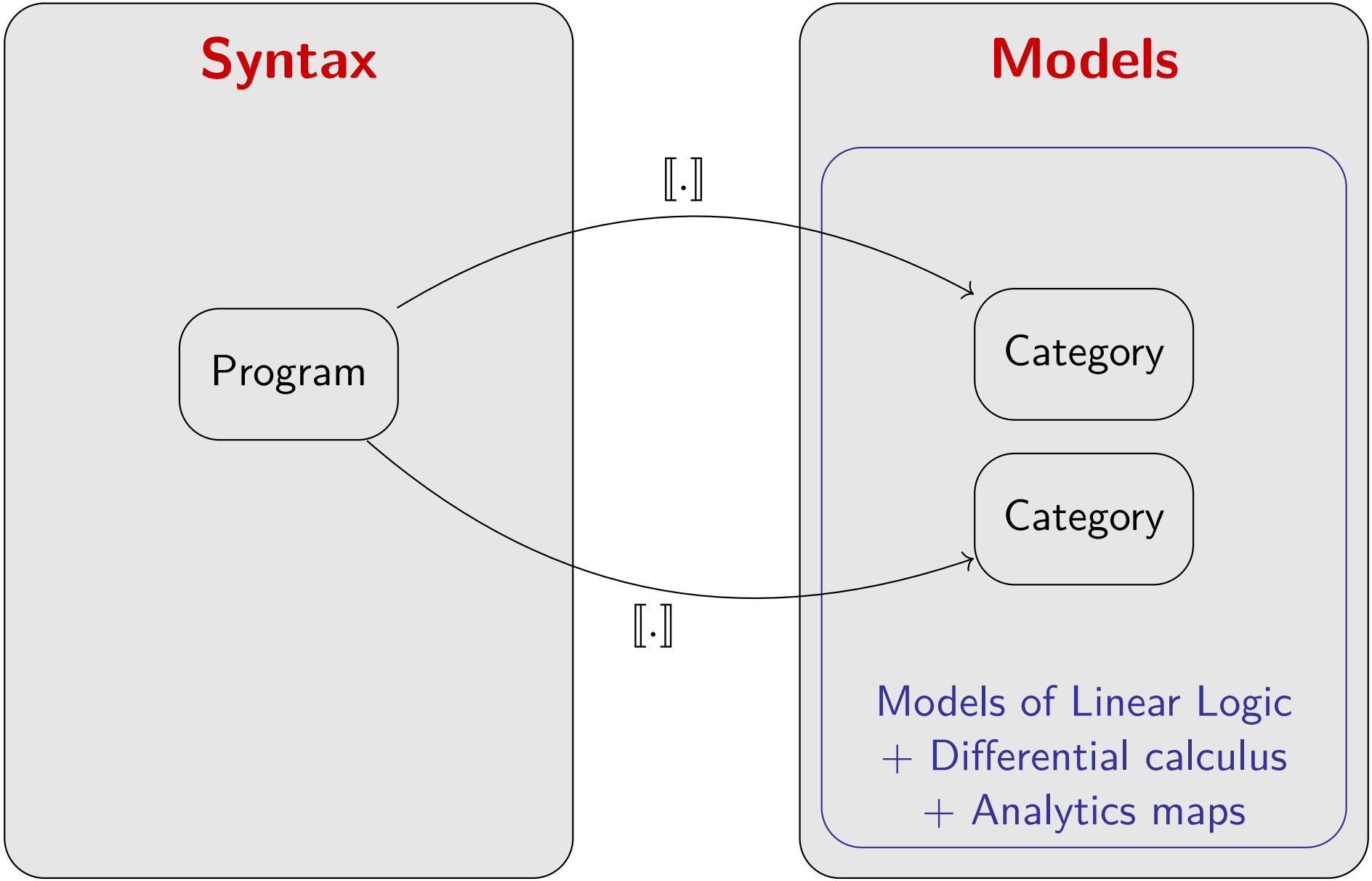
Analytic map

$$f(x+u) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(x) \cdot (\underbrace{u, \ldots, u}_{k \text{ times}})$$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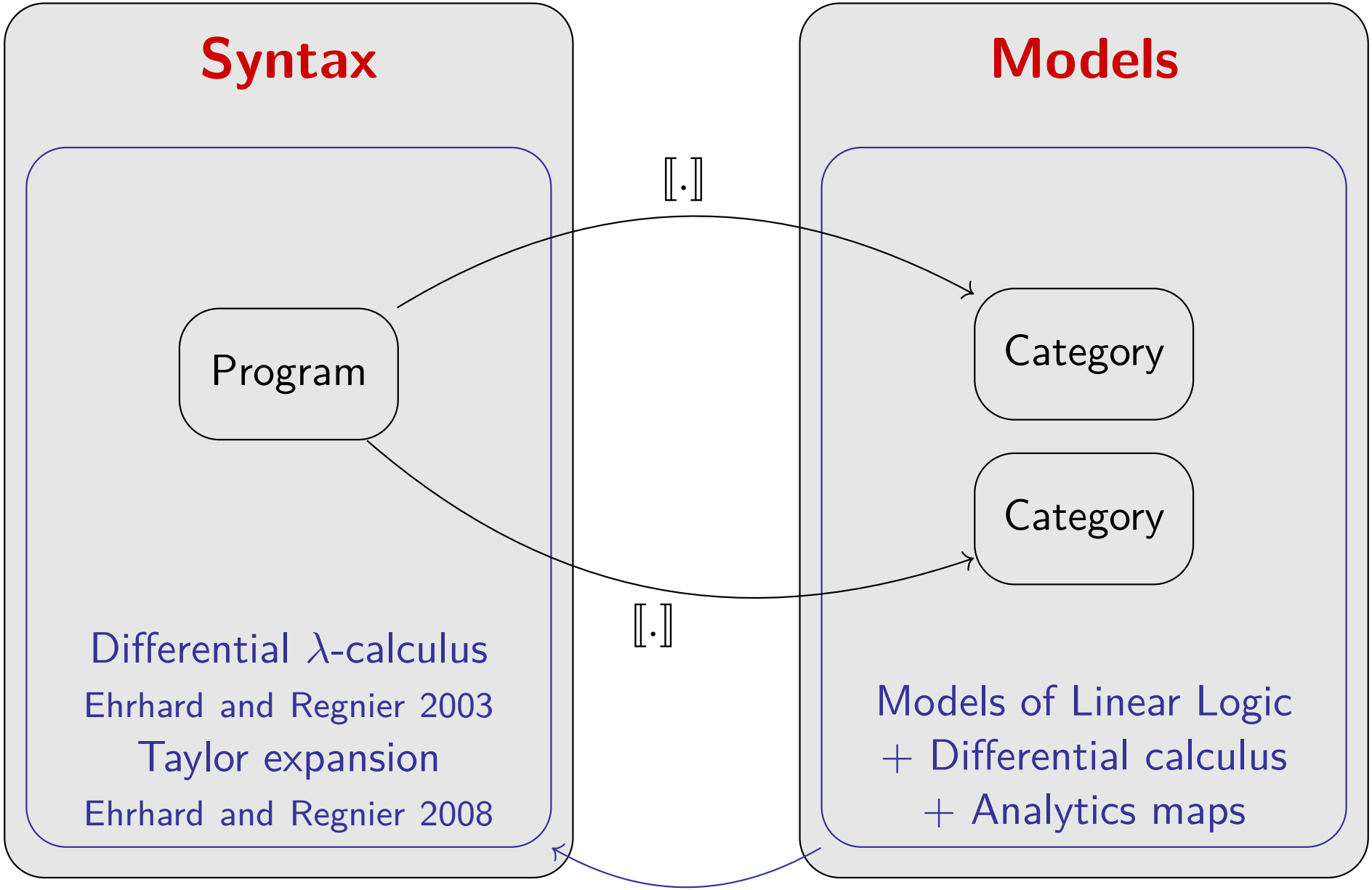
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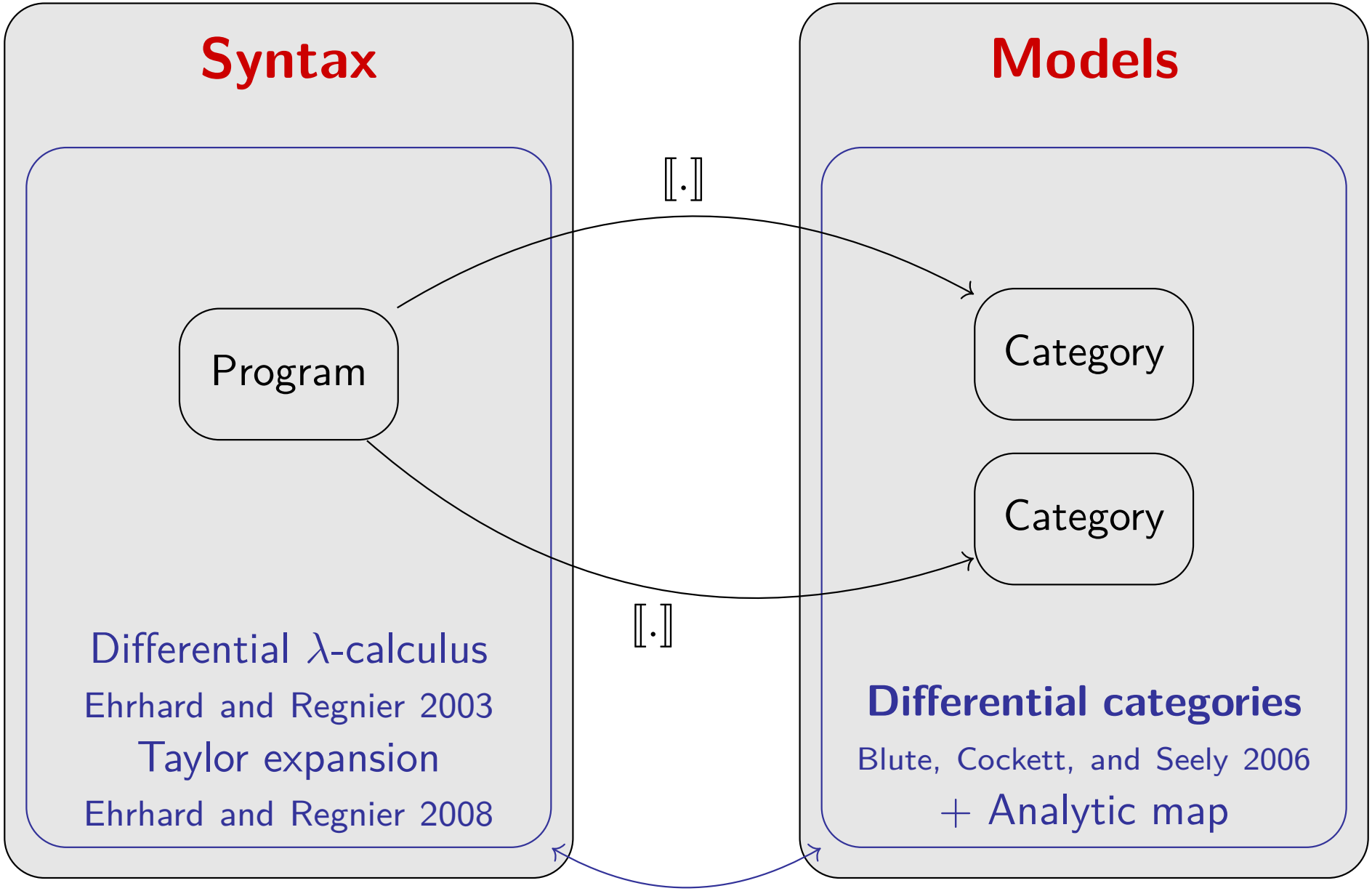
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Differential categories
Blute, Cockett, and Seely 2006
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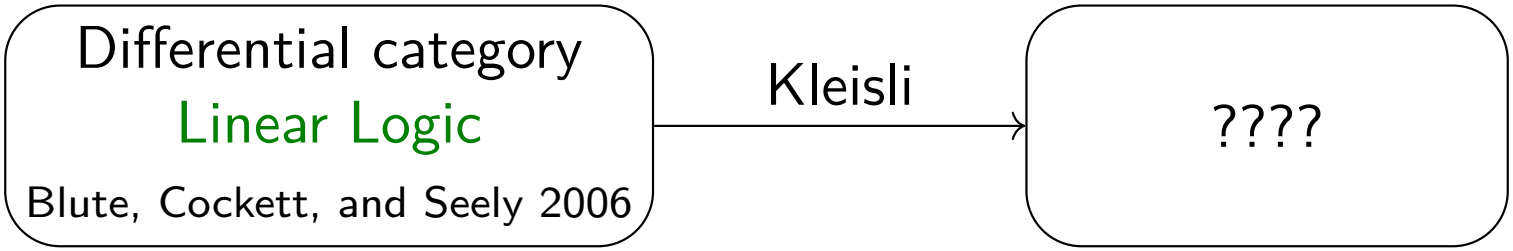
The birth of categorical differentiation



The birth of categorical differentiation

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Cartesian differential category

Blute, Cockett, and Seely 2009

The birth of categorical differentiation

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- When closed: models of the differential λ-calculus (Bucciarelli, Ehrhard, and Manzonetto 2010) and the syntactical Taylor expansion (Manzonetto12)
- Related to the semantics of automated differentiation
- Algebraic theory of differentiation in maths

The problem of sums

Sums, sums everywhere...

"Leibniz" rule: if $f : E \times F \to G$

$$f'(x,y) \cdot (u,v) = \frac{\partial f}{\partial x}(x,y) \cdot u + \frac{\partial f}{\partial y}(x,y) \cdot v$$

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If we want Taylor expansion

...We allow **countable** sums and non-determinism.

The problem

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- Some lack finite sums: coherence spaces, probabilistic coherence spaces

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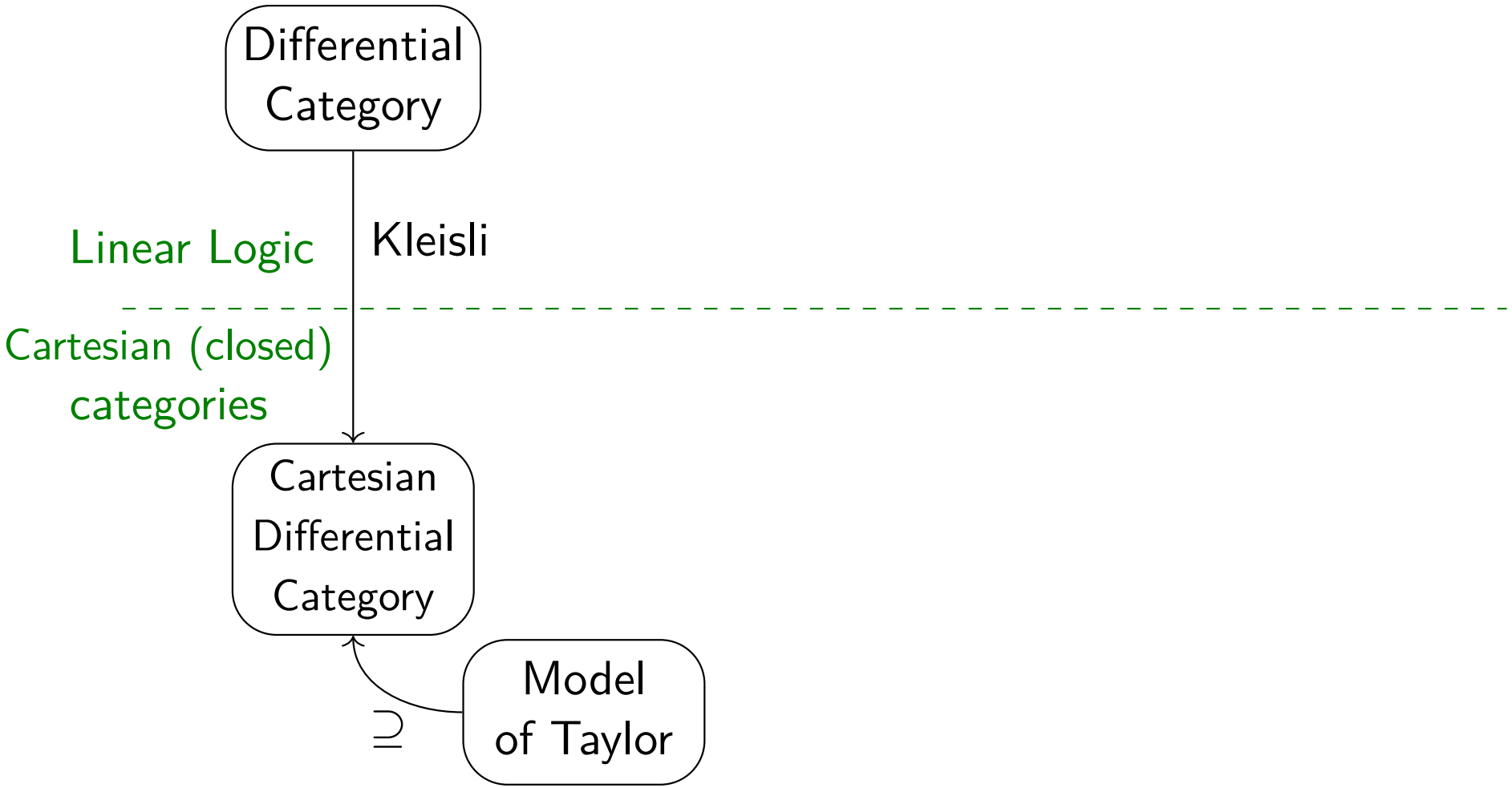
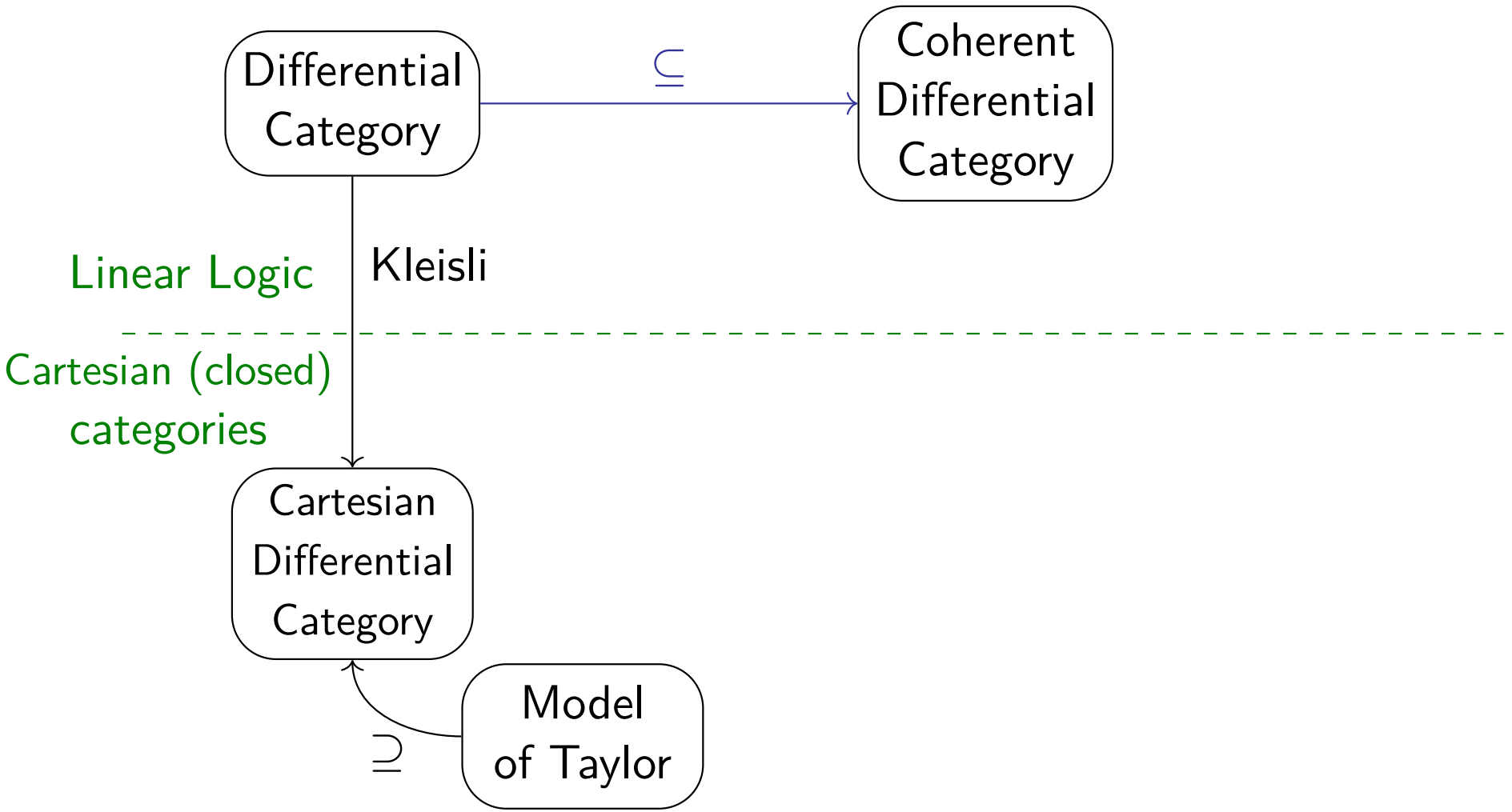
Objective of the thesis

We want a theory of differential linear logic and Taylor expansion with partial sums.


Differential Category
Linear Logic
Kleisli
Cartesian (closed) categories
Cartesian Differential Category
Model of Taylor
⊇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Coherent Differential Category
Linear Logic
Kleisli
Cartesian (closed) categories
Cartesian Differential Category
Model of Taylor
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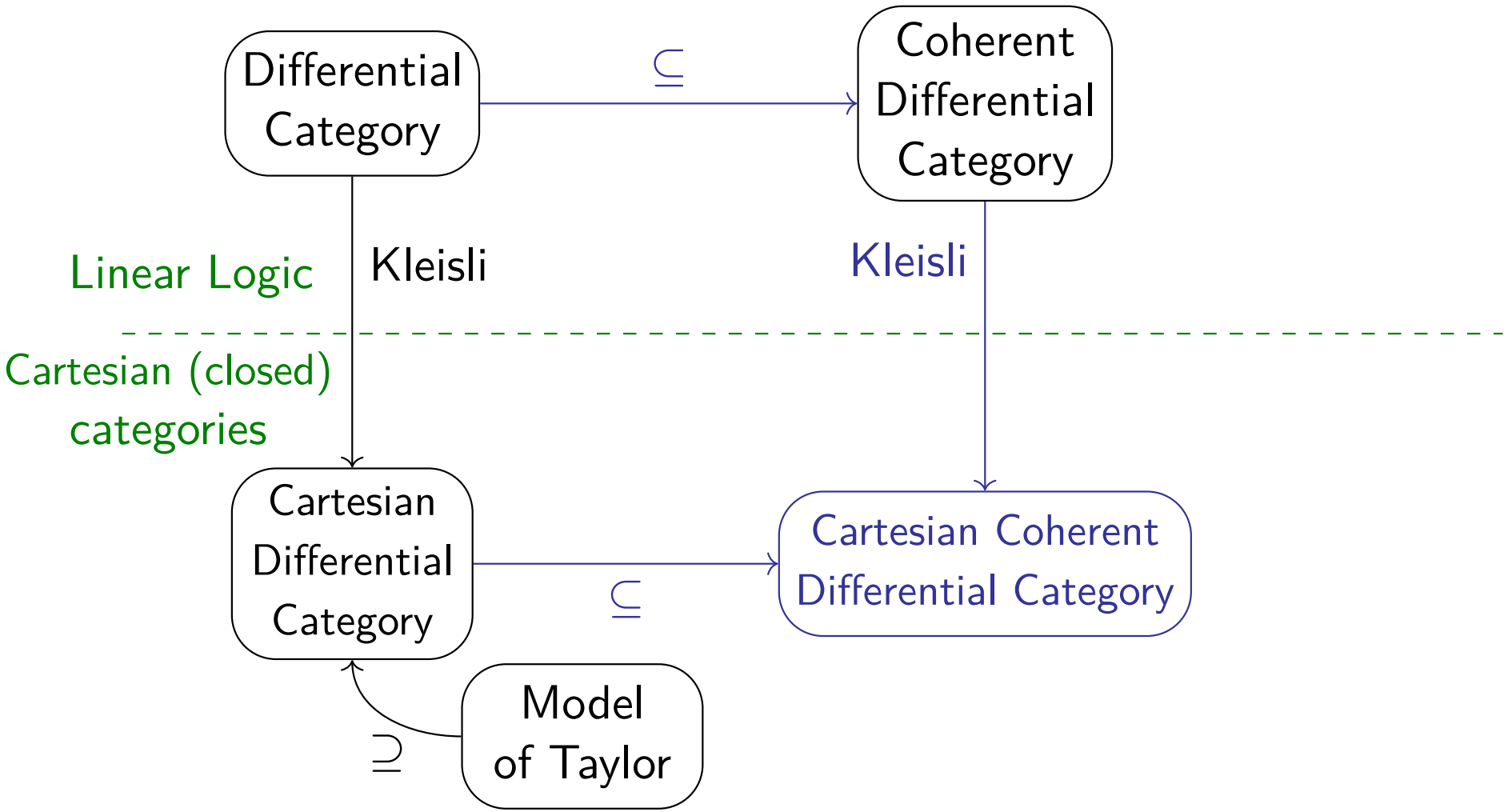
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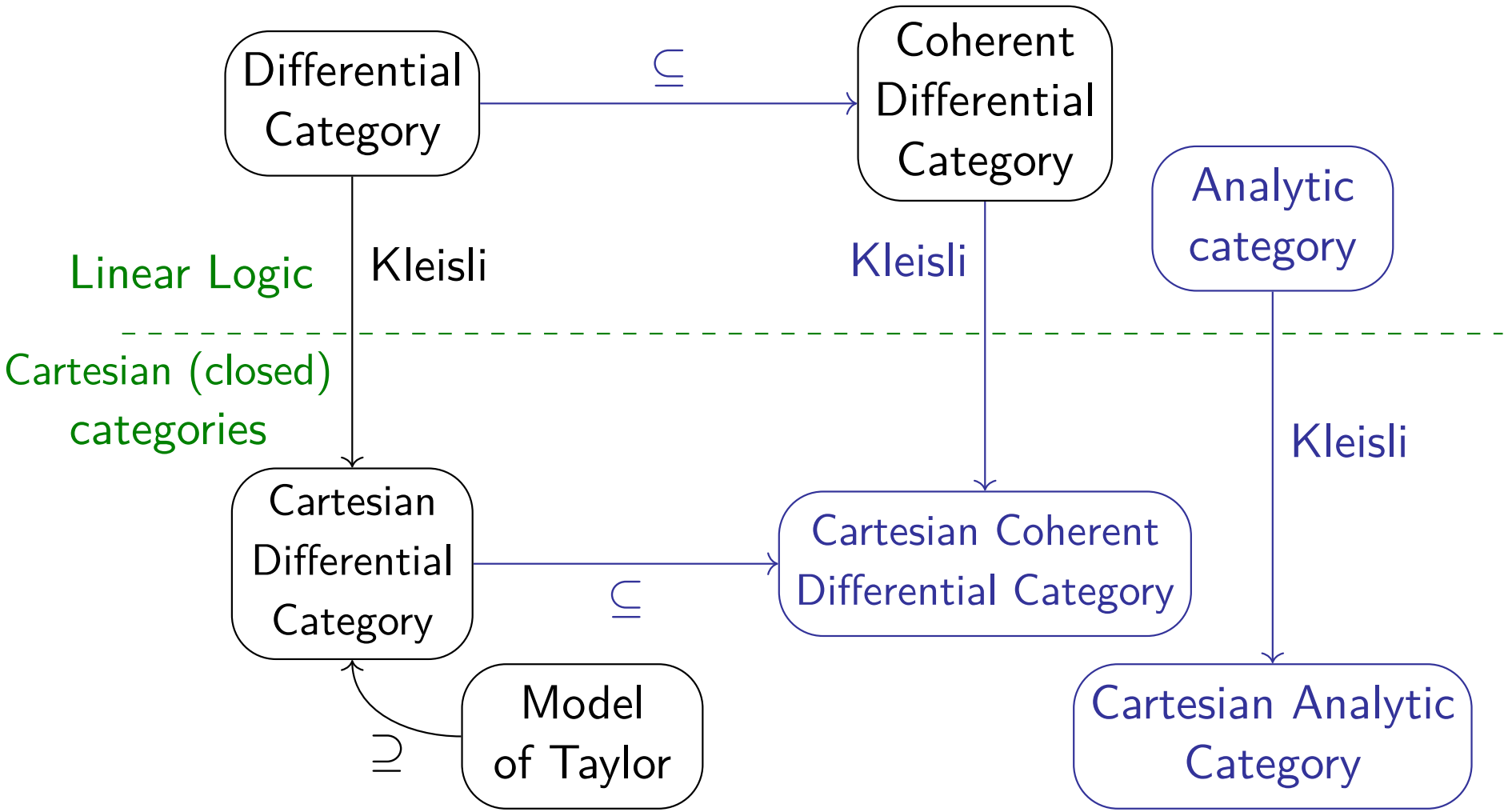
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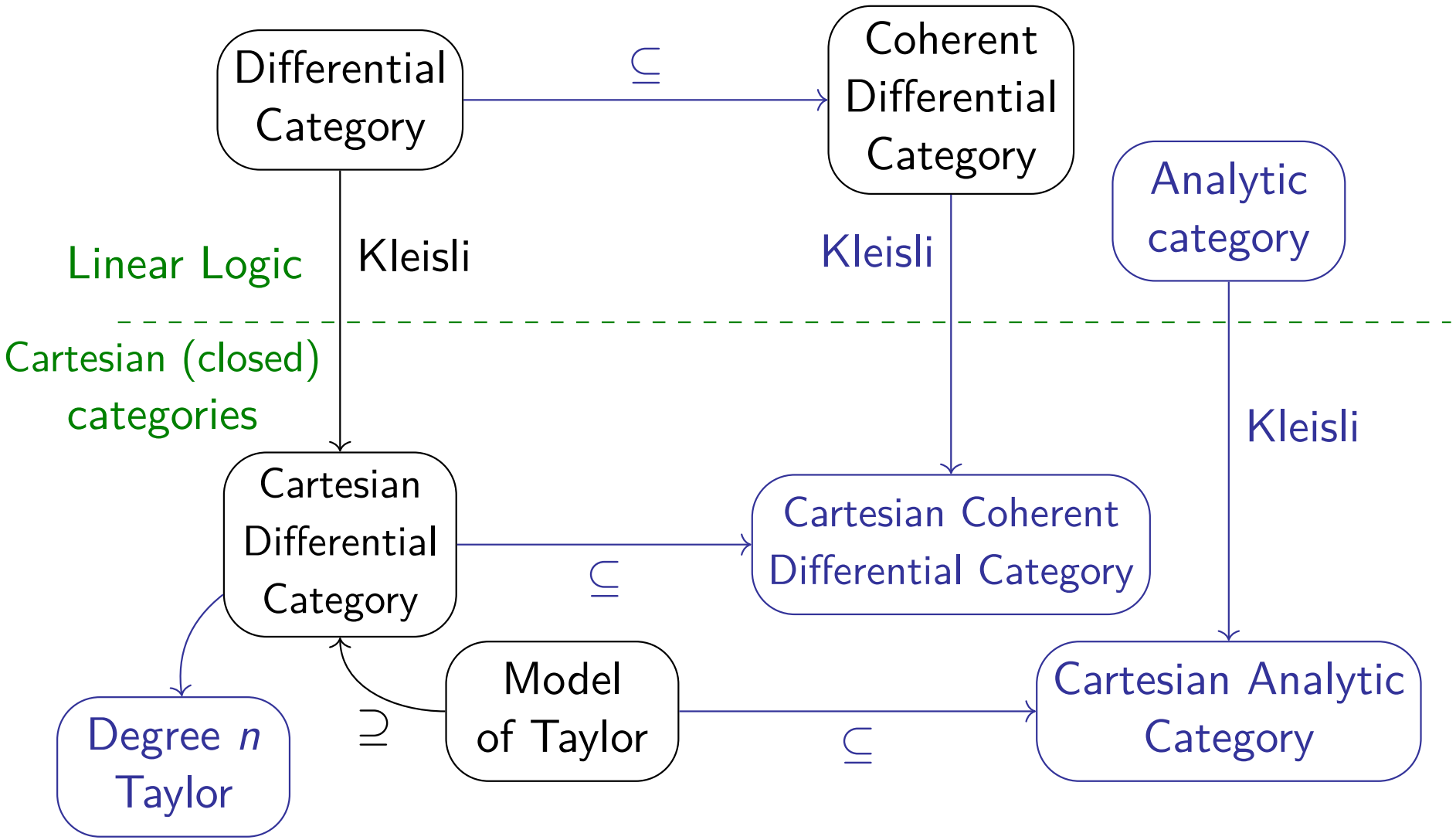
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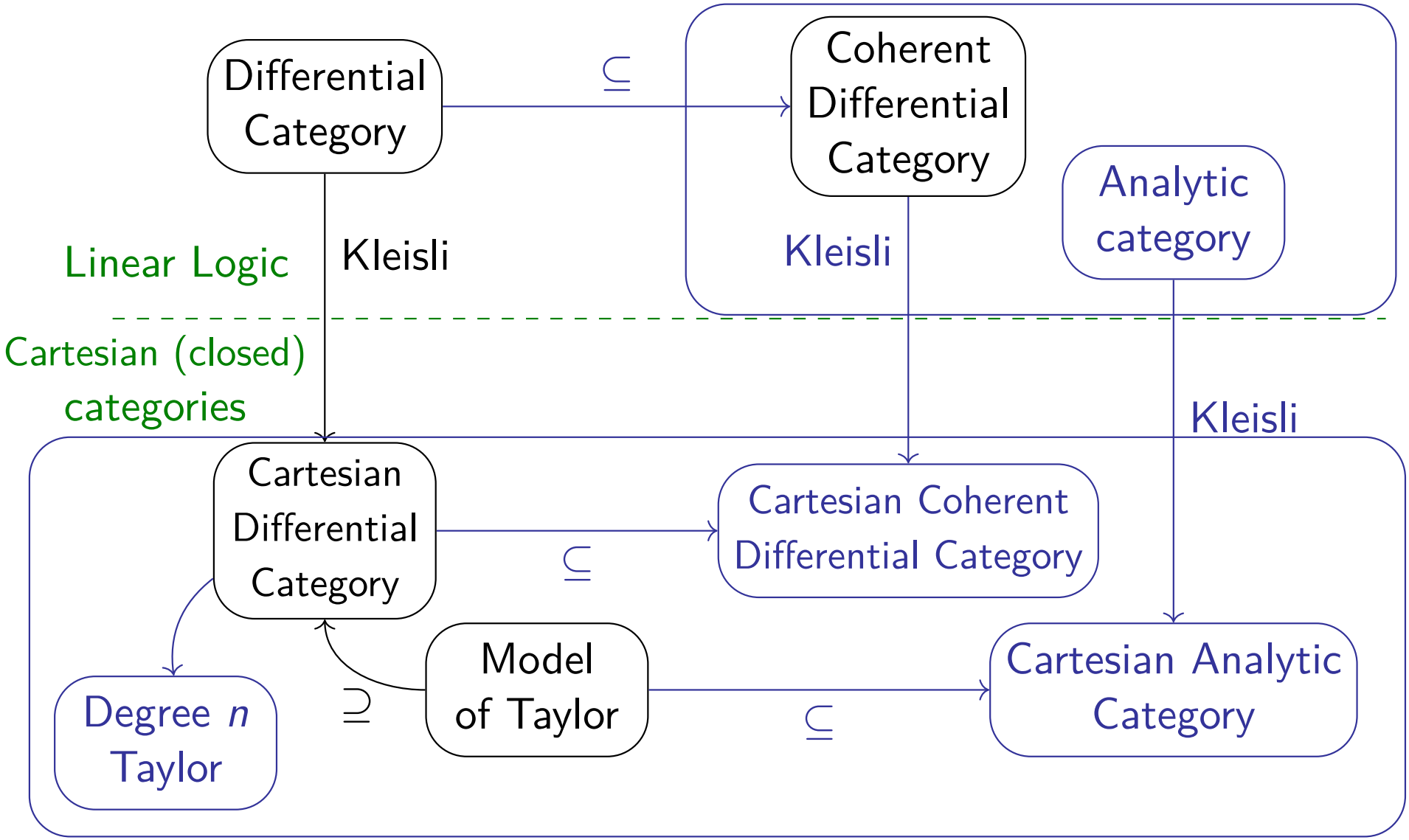
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⊇
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Cartesian Analytic Category

Compositional Taylor expansion in additive and non-additive models of Linear Logic

We start with

the derivative

Cartesian differential category

Cartesian differential category

Category + sum between morphisms + **derivative operator**

$$\frac{f : X \to Y}{\mathrm{D}f : X \times X \to Y}$$

+ equations on D (properties of derivatives).

Example: the category **Smooth**

Morphisms are smooth functions $f, g : X \to Y$.

$$\mathrm{D}f : (x, u) \mapsto f'(x) \cdot u$$

Fixing the chain rule

D must satisfy the chain rule.

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$$

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Tangent bundle

$$\frac{f : X \to Y}{\mathsf{T}f : X \times X \to Y \times Y} \qquad \mathsf{T}f(x,u) = \Big(f(x), f'(x) \cdot u\Big)$$

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Dual number intuition

$$(x,u) \in \mathsf{T}X \iff \text{polynomial } x + u\varepsilon \text{ with } \varepsilon^2 = 0$$

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$$f(x + u\varepsilon) = f(x) + \varepsilon f'(x) \cdot u \ + O(\varepsilon^2)$$

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$$f(x + u\varepsilon) = f(x) + \varepsilon f'(x) \cdot u \; + \cancel{O(\varepsilon^2)} \qquad \textbf{push forward}$$

$\mathsf{T}f(x,u) = (f(x), f'(x) \cdot u)$: degree 1 Taylor expansion

Let us do the same

for Taylor expansion

A degree 2 Taylor expansion

Tangent bundle

$$\mathsf{T}X = X \times X \qquad \frac{f : X \to Y}{\mathsf{T}f : \mathsf{T}X \to \mathsf{T}Y}$$

Dual number intuition

$$(x, u) \in \mathsf{T}X \iff \text{polynomial } x + u\varepsilon \text{ with } \varepsilon^2 = 0$$

$\mathsf{T}f(x, u)$: **push forward** of polynomials

A degree 2 Taylor expansion

Degree 2 Taylor bundle

$$\mathrm{T}_2 X = X \times X \times X \qquad \frac{f : X \to Y}{\mathrm{T}_2 f : \mathrm{T}_2 X \to \mathrm{T}_2 Y}$$

Higher order dual numbers intuition

$$(x, u, v) \in \mathrm{T}_2 X \iff \text{polynomial } x + u\varepsilon + v\varepsilon^2 \text{ with } \varepsilon^3 = 0$$

$\mathrm{T}_2 f(x, u, v)$: **push forward** of polynomial.

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$$f(x + u\varepsilon + v\varepsilon^2)$$
$$= f(x) + f'(x) \cdot (u\varepsilon + v\varepsilon^2) + \frac{1}{2} f^{(2)}(x) \cdot (u\varepsilon + v\varepsilon^2, u\varepsilon + v\varepsilon^2) + \cancel{O(\varepsilon^3)}$$

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$$\begin{aligned} & f(x + u\varepsilon + v\varepsilon^2) \\ &= f(x) + f'(x) \cdot (u\varepsilon + v\varepsilon^2) + \frac{1}{2} f^{(2)}(x) \cdot (u\varepsilon + v\varepsilon^2, u\varepsilon + v\varepsilon^2) + \cancel{O(\varepsilon^3)} \\ &= f(x) + \varepsilon f'(x) \cdot u + \varepsilon^2 \left(\frac{1}{2} f^{(2)}(x) \cdot (u, u) + f'(x) \cdot v \right) + \cancel{\varepsilon^3(\ldots)} \end{aligned}$$

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A compositional Taylor expansion

Degree n Taylor bundle

$$\mathsf{T}_n X = X \times X^n \qquad \frac{f : X \to Y}{\mathsf{T}_n f : \mathsf{T}_n X \to \mathsf{T}_n Y}$$

Higher order dual numbers intuition

$$(x, u_1, \ldots, u_n) \in \mathsf{T}_n X \iff \text{polynomial } x + \sum_{i=1}^{n} u_i \varepsilon^i \text{ with } \varepsilon^{n+1} = 0$$

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$$\mathsf{T}_n f(x, u, 0, \ldots, 0) = \left(\frac{1}{j!} f^{(j)}(x) \cdot (u, \ldots, u) \right)_{j=0}^{n}$$

A compositional Taylor expansion

Faá di Bruno formula: non-compositional (and scary) ☹

$$(g \circ f)^{(n)}(x) = \sum_{\{I_1,\dots,I_k\} \text{ partition of } [\![1,n]\!]} g^{(k)}\left(f(x), f^{(I_1)}(x), \dots, f^{(I_k)}(x)\right)$$

with $f^{(\{i_1,\dots,i_l\})}(x) : (u_1, \dots, u_n) \mapsto f^{(I)}(x)(u_{i_1}, \dots, u_{i_l})$

Theorem 1. (Chapter 6)

$$\mathsf{T}_n(g \circ f) = \mathsf{T}_n g \circ \mathsf{T}_n f$$ ☺

Categorical structure of T_n

T_n exists in all **cartesian differential categories**

Theorem 2. (Chapter 5)

T_n *is a functor* $\iff$ *Chain rule on* D

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T_n *is a monad*
$\mathsf{z} : \mathsf{Id} \Rightarrow \mathsf{T}_n$, $\theta : \mathsf{T}_n\mathsf{T}_n \Rightarrow \mathsf{T}_n$

Distributive law $\mathsf{c} : \mathsf{T}_n\mathsf{T}_n \Rightarrow \mathsf{T}_n\mathsf{T}_n$ $\iff$ *On axioms on* D

Natural transformation
$\mathsf{I} : \mathsf{T}_n \Rightarrow \mathsf{T}_n\mathsf{T}_n$

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$$\pi_n \circ \theta = \sum_{i+j=n} \pi_i \circ \pi_j$$

Proof by... computation ?

Remember

$$\mathsf{T}_n f(x, u_1, \ldots, u_n) = \left(\sum_{k=1}^{j} \sum_{\substack{i_1, \ldots, i_k \text{ s.t.} \\ i_1 + \cdots + i_k = j}} \frac{1}{k!} f^{(k)}(x) \cdot (u_{i_1}, \ldots, u_{i_k}) \right)_{j=0}^{n}$$

Explicit computation ? ☹

Category theory proof ☺

Also works for $n = \infty$

$$\mathsf{T}_\omega X = X \times X \times \cdots$$

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One last ingredient

Morphisms are analytic

$$f(x+u) = \sum_{k \in \mathbb{N}} f^{(k)}(x) \cdot (u, \ldots, u)$$

Also works for $n = \infty$

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$$f(x+u) = \sum_{k \in \mathbb{N}} f^{(k)}(x) \cdot (u, \ldots, u)$$

If and only if

$$f \circ \sigma = \sigma \circ \mathsf{T}_\omega f$$

where $\sigma(x_0, x_1, x_2, \ldots) = \sum_{i=0}^{\infty} x_i$

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T_ω is a bimonad

Compositional Taylor expansion in additive and non-additive models of Linear Logic

Partial sum: change the action on objects

Total sums

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Partial sums

Partial sum on morphisms: Partial Commutative Monoid (PCM)

$$\mathsf{T}_\omega X = \left\{ (x_i)_{i \in \mathbb{N}} \mid \sum_{i \in \mathbb{N}} x_i \text{ is defined} \right\}$$

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Formally: $\mathsf{T}_\omega X$ is a **summability structure**

- Projections $\pi_i : \mathsf{T}_\omega X \to X$ jointly monic
- Sum operation $\sigma : \mathsf{T}_\omega X \to X$
- $(f_i)_{i\in\mathbb{N}}$ is summable if and only if there is f such that $\pi_i \circ f = f_i$

- Total sums: $\mathsf{T}_\omega X = X \times X \times \cdots$
- Partial sums: $\mathsf{T}_\omega X$ is a summability structure

Same categorical structure: T_ω is a bimonad.

$$\pi_n \circ \theta = \sum_{i+j=n} \pi_i \circ \pi_j$$

Compositional Taylor expansion in additive and non-additive models of Linear Logic

Let $\mathcal{L}$ be a model of LL.

- Partial sums on morphism: Partial Commutative Monoid (PCM)
- Summability structure SX

$$\mathrm{S}X = \{(x_i)_{i\in\mathbb{N}} \mid \sum_{i=0}^{\infty} x_i \text{ defined } \}$$

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No derivative yet

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Taylor expansion is in the Kleisli category $\mathcal{L}_!$

Taylor expansion: bimonad T in $\mathcal{L}_!$ that **extends** S

1)
$$\begin{array}{ccc} \mathcal{L} & \xrightarrow{\mathsf{S}} & \mathcal{L} \\ {\scriptstyle \mathsf{Der}}\downarrow & & \downarrow{\scriptstyle \mathsf{Der}} \\ \mathcal{L}_! & \xrightarrow[\mathsf{T}]{} & \mathcal{L}_! \end{array}$$

No derivative yet

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2) The bimonad structure on T is the same as the bimonad structure on S

This is the same

- Bimonad T that extends S
- Natural transformation $\partial : !S \Rightarrow S!$
  - **Distributive law** between comonad !_ and functor S
  - Bimonad structure on S: **morphisms of distributive laws**

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$$\begin{array}{ccccc} !SS & \xrightarrow{\partial S} & S!S & \xrightarrow{S\partial} & SS! \\ {\scriptstyle !\theta}\downarrow & & & & \downarrow{\scriptstyle \theta!} \\ !S & & \xrightarrow[\partial]{} & & S! \end{array}$$

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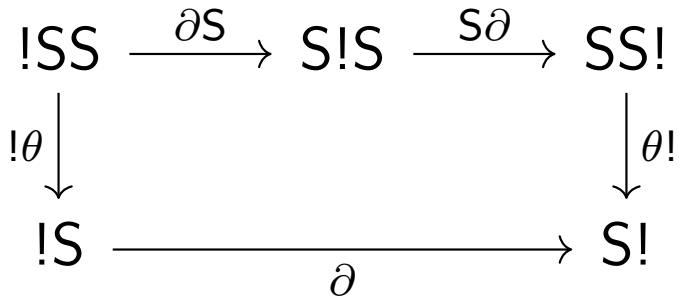

Theorem 3. (Chapter 9)

*The Kleisli category $\mathcal{L}_S$ of the **monad** S is a model of LL.*

Time to build concrete models!

The representable theory

Introduced by Ehrhard 2023 for coherent differentiation.

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$$\mathsf{S}X = \mathbb{D} \multimap X \quad \text{with} \quad \mathbb{D} = \bigwith_{i \in \mathbb{N}} 1$$

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What does that means?

$$\text{Injection } \iota_i = (\overbrace{0, 0, \ldots, 0}^{i-1}, 1, 0, \ldots) \in \mathbb{D}$$

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Epicity assumption: An element $s \in \mathbb{D} \multimap X$ is the same as $(s(\iota_i))_{i \in \mathbb{N}}$

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Epicity assumption: An element $s \in \mathbb{D} \multimap X$ is the same as $(s(\iota_i))_{i \in \mathbb{N}}$

$$\sum_{i \in \mathbb{N}} s(\iota_i) = s(1, 1, 1, \ldots)$$

A recipe for guessing sums

In coherence spaces

A family of clique is summable if

- The union is a clique
- The intersection are pairwise disjoints

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In coherence spaces

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In probabilistic coherence spaces

A family of sub probability distributions is summable if the sum is also a sub probability distribution.

Lots of models !

Theorem 4. (Chapter 10)

Every Lafont model of Linear Logic such that $\mathbb{D} \multimap \_$ is well-defined has a Taylor expansion.

Proof: distributive laws, adjunctions, **Mates**

Lots of models !

Theorem 4. (Chapter 10)

Every Lafont model of Linear Logic such that $\mathbb{D} \multimap \_$ is well-defined has a Taylor expansion.

Proof: distributive laws, adjunctions, **Mates**

- Relational models
- Weighted relational models
- Coherence spaces (Girard, non uniform, weak)
- Probabilistic coherence spaces
- WIP: Köthe spaces and finiteness spaces?

To conclude

Conclusion

A **compositional** theory of Taylor expansion

- With total or partial sums
- In cartesian (closed) categories or in models of LL

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- In cartesian (closed) categories or in models of LL

Future work

Linear Logic flavour:

- A syntax of coherent Taylor expansion
- More models Köthe spaces, Finiteness spaces
- Alternative bimonads $\mathbb{D} \multimap \_$

Cartesian (closed) categories flavour:

- Relate with higher order automated differentiation
- Axiomatization in tangent categories: jet bundles

1. Denotational semantics
2. Linear Logic
3. Compositional Taylor expansion
4. ... in Linear Logic
5. The end


Girard, Jean-Yves (1987). "Linear Logic". In: Theoretical Computer Science 50, pp. 1–102. DOI: 10.1016/0304-3975(87)90045-4. URL: https://doi.org/10.1016/0304-3975(87)90045-4.


Blute, R., Robin Cockett, and R. Seely (Dec. 2006). "Differential categories". In: Mathematical Structures in Computer Science 16, pp. 1049–1083. DOI: 10.1017/S0960129506005676.


Ehrhard, Thomas and Laurent Regnier (2003). "The differential lambda-calculus". In: Theoretical Computer Science 309.1-3, pp. 1–41.


— (2008). "Uniformity and the Taylor expansion of ordinary lambda-terms". In: Theoretical Computer Science 403.2, pp. 347–372. ISSN: 0304-3975. DOI: https://doi.org/10.1016/j.tcs.2008.06.001. URL: https://www.sciencedirect.com/science/article/pii/S0304397508004064.


Bucciarelli, Antonio, Thomas Ehrhard, and Giulio Manzonetto (2010). "Categorical Models for Simply Typed Resource Calculi". In: Electronic Notes in Theoretical Computer Science 265. Proceedings of the 26th Conference on the Mathematical Foundations of Programming Semantics (MFPS 2010), pp. 213–230. ISSN: 1571-0661. DOI: https://doi.org/10.1016/j.entcs.2010.08.013. URL: https://www.sciencedirect.com/science/article/pii/S1571066110000927.


Blute, R., Robin Cockett, and R. Seely (Jan. 2009). "Cartesian differential categories". In: Theory and Applications of Categories 22, pp. 622–672.


Ehrhard, Thomas (2023). “Coherent differentiation”. In: Mathematical Structures in Computer Science, pp. 1–52. DOI: 10.1017/S0960129523000129.